

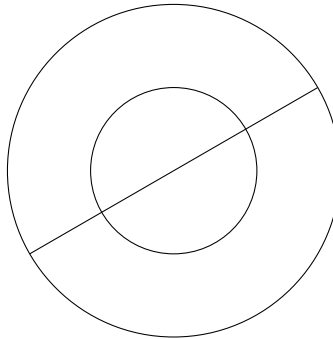
An Exploration of Probability Theory and Financial Mathematics

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Part 1: Bertrand's Paradox

Joseph Bertrand was a French mathematician, most famous for his work in probability theory. In 1889, he proposed a problem in which a circle of radius 1 is inscribed within another circle of radius 2, with the same centre. He questioned what the probability is that a randomly drawn chord from any point on the outer circle will intersect the inner circle, as shown in *figure 1.1*.

figure 1.1

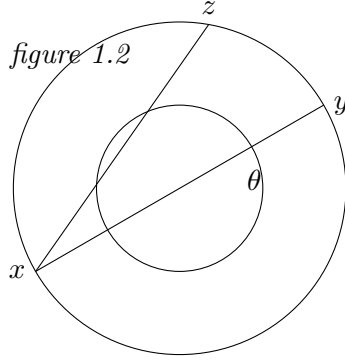


Classically, we can approach this problem in two ways. Let us first define the event of the chord intersecting with the inner circle as event A . This event can only occur if the midpoint of the chord lies within the inner circle. This logically follows from the fact that a chord tangential to that circle would meet the circumference at its midpoint, such that any rotation of this chord about its point on the circumference of the outer circle would result in the midpoint now being inside the inner circle. Since the midpoint can be anywhere inside the inner circle, and is also, therefore, inside the outer circle, we can consider probability to correspond to the areas of space in which the event can occur, and define $P(A)$ as the following,

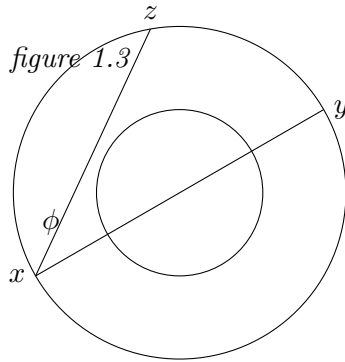
$$P(A) = \frac{\text{Area of inner circle}}{\text{Area of outer circle}} = \frac{\frac{1}{2}\pi \times 1^2}{\frac{1}{2}\pi \times 2^2} = \frac{1}{4}$$

Alternatively, we can consider any chord as a variation of the diameter of the outer circle as a line, $\overline{xy}$. Suppose a diameter which starts from point x . Rotating this diameter by an angle of θ , about the point x will result in a new chord of the outer circle, $\overline{xz}$, under the following condition, which maintains the chord's presence within the outer circle.

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$



To find the maximum angle that the diameter of the circle can rotate such that the new chord, $\overline{xz}$, intersects with the inner circle, we can use the angle between the chord and the diameter when the chord is a tangent to the inner circle. Let ϕ be the angle between the diameter and the new chord. Since $\overline{xz}$ is a tangent, the distance between the point where it touches the inner circle and the midpoint of the diameter is simply the radius of the inner circle, 1. From this, we can form a right-angled triangle with which we can apply trigonometric algebra to find the range of angles encompassing the event A .



$$\begin{aligned} \therefore \sin(\phi) &= \frac{1}{2} \\ -\frac{\pi}{6} &< \phi < \frac{\pi}{6} \end{aligned}$$

We can now find the probability of event A to be all the possible rotations of the diameter that will produce a chord that intersects with the inner circle, divided by all of those which produce a general chord of the outer circle.

$$P(A) = \frac{\frac{\pi}{6}}{\frac{\pi}{2}} = \frac{1}{3}$$

Indeed, we have reached a contradiction. The probability that event A is satisfied cannot be both $\frac{1}{3}$ and $\frac{1}{4}$. However, the mistake is not in the logic of either method, but is instead in the fact that we have modelled an infinite sample space as a finite one. There are an endless number of points where the midpoint of the chord can lie within the circle, and, likewise, an infinite number of different values for θ and ϕ . This poses the question, how can we ensure that our application of probabilistic methods is accurately modelled?

Part 2: Defining Probability, Finite Sample Spaces

Fundamentally, probability is a tool used to understand randomness. A variable can be defined as random if we have no prior knowledge of the outcome of that variable. For example, when flipping a coin, we can either get heads or tails, but cannot know what the outcome will be before we have flipped the coin. In this case, the set of all possible outcomes, or the sample space, Ω , is finite.

When working with finite sample spaces, an event in that space is any subset of Ω . Let A be an event in Ω .

$$A \subset \Omega$$

The probability of A , then, is the number of elements in A , divided by the number of elements in Ω .

$$P(A) = \frac{\text{card}(A)}{\text{card}(\Omega)} \quad \text{where } \text{card}() \text{ is the cardinality of the set, or the number of elements contained in the set.}$$

It is true that this definition of probability is valid in many cases and has many applications, however, it is limited by 4 key assumptions:

- All outcomes are mutually exclusive
- All outcomes are exhaustive
- All outcomes are equally likely
- The sample space is finite

Part 3: Defining Probability, Infinite Sample Spaces

Finite and infinite sample spaces are primarily distinguished by what we can define as an event. In the world of infinite sample spaces, we cannot consider all possible subsets of our sample space, Ω , as events.

The rules of infinite sample space probability were rigorously defined in 1933 by Russian mathematician Andrey Kolmogorov when he published his *Basic Concepts of Probability Theory*. He stated that we can define a collection of subsets, U , of Ω , if the following axioms are met:

$$P(\emptyset) = 0$$

The probability of the empty set is 0

$$P(\Omega) = 1$$

The probability of the sample space is 1

$$A_1, A_2, A_3, \dots, A_k, \dots \in U, \quad \text{then } P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{k=1}^{\infty} P(A_k)$$

If we have an infinite sequence of events that are mutually disjoint (where no two sets have any elements in common), then the infinite union of all of the sets is equal to the sum of the probabilities.

From these axioms, we can derive the identity,

$$P(A^c) = 1 - P(A) \quad \text{where } A^c \text{ is the non-occurrence of } A$$

Figure 3.1 displays this identity clearly.

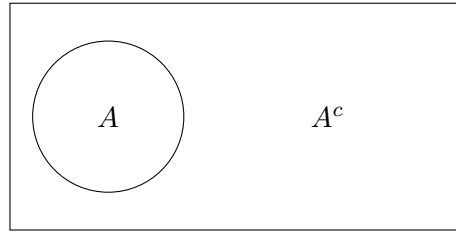


figure 3.1

Kolmogorov proceeded to redefine probability as a function, P , which maps U , to a number between 0 and 1. The combination of the probability measure, P , the sample space, Ω , and the subset of events, U , forms what is called the probability space.

$$(\Omega, U, P)$$

Part 4: Buffon's Needle Problem

Using our developed understanding of probability, we can tackle a new problem. In 1733, Georges-Louis Leclerc, Comte de Buffon, imagined drawing a set of parallel lines on the floor, 2cm apart from each other, and randomly throwing a needle of length 1cm on the floor. He wanted to find the probability that the needle touched one of the lines.

The first step to solving this problem is to construct our probability space. If we imagine that the needle does touch the line, we can call the shortest distance from the end of the needle to the line on the floor, h , and the angle that is made between the floor and the line, θ .

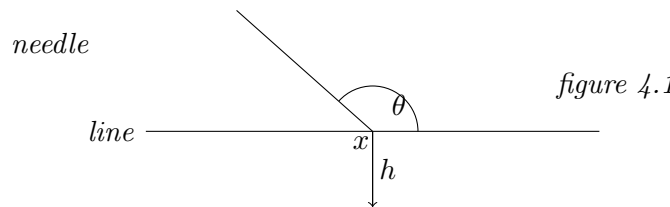


figure 4.1

For the needle to touch the line, the distance x must be in-between 0 and $\frac{1}{2}$, since the needle is of length 1cm.

$$\therefore 0 \leq \frac{h}{\sin(\theta)} \leq \frac{1}{2} \quad \text{because } x = \frac{h}{\sin(\theta)}$$

Once again, let us define the event where the needle touches the line as B . We can say that for all

$$0 \leq \theta \leq \frac{\pi}{2}, \quad h \leq \frac{1}{2} \sin(\theta)$$

Now, $P(B) = \frac{\text{area of } B}{\text{area of } \Omega}$ where Ω is the entire sample space.

The area of B is the area under $y = \frac{1}{2} \sin(\theta)$, *figure 4.2*, from 0 to $\frac{\pi}{2}$, since those are the limits of θ .

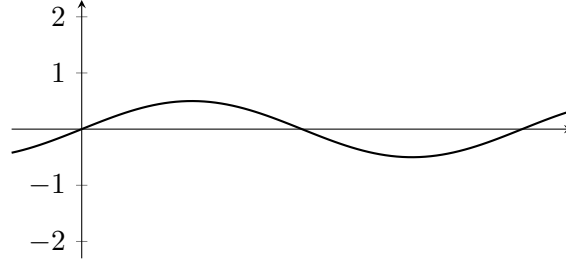


figure 4.2

This area is given by $\int_0^{\pi/2} \frac{1}{2} \sin(\theta) d\theta$ which evaluates to $\frac{1}{2}$.

Finally, the area of Ω is simply the product of the two upper limits of θ and h , which is $1 \times \frac{\pi}{2} = \frac{\pi}{2}$.

$$\text{Thus, } P(B) = \frac{\frac{1}{2}}{\frac{\pi}{2}} = \frac{1}{\pi}$$

An approximation of this result was published by Buffon in 1777, however, a rigorous proof was formulated in 1930 by Kolmogorov, based on his formalisation of modern probability theory.

Part 5: Law of Total Expectation, Markov's Inequality and Chebyshev's Lemma

Suppose a small country. 90% of its population earn an average wage of £30,000 per year, while 10% earn an average of £230,000 per year. To determine the average wage of the entire population, we must weight each value for wage with regards to its probability.

To determine the average wage: $\frac{9}{10} \times 30,000 + \frac{1}{10} \times 230,000$

Given our information on the distribution of wages within the population, we have determined that the probability a randomly chosen person earns less than some value, t where $30,000 < t < 230,000$, is $\frac{9}{10}$, and the probability someone earns more than, or at least, t , is $\frac{1}{10}$. Mathematically, we can denote these probabilities as,

$$P(X < t) = \frac{9}{10} \quad \text{and} \quad P(X \geq t) = \frac{1}{10}$$

where X is the non-negative random variable, wage (£) per year.

We can also refer to the given wages that correspond to each probability as conditional expectations of X . $E(X)$ represents the expected value of X , more commonly referred to as the mean, μ :

$$E(X \mid X < t) = 30,000 \quad \text{and} \quad E(X \mid X \geq t) = 230,000$$

Thus, from our original calculation of the mean, or expectation of X , we can generally describe $E(X)$ as:

$$E(X \mid X < t) \times P(X < t) + E(X \mid X \geq t) \times P(X \geq t) = E(X)$$

Between 1940 and 1960, this result was explicitly formalised and discussed as the law of total expectation, in the context of Kolmogorov's axioms, published in the 1930s. The law of total expectation directly leads us to Markov's inequality, which investigates how to bound the probabilities of a variable and is key in the exploration of stochastic processes, a concept which will be further explained shortly.

Now, using the result above, consider fixing the two probabilities such that they are constant. We can see that there is no upper limit for $E(X \mid X \geq t)$, as it is not bound by t , meaning that there is no upper limit for $E(X)$: $E(X \mid X \geq t) \rightarrow \infty$, $E(X) \rightarrow \infty$. $E(X)$ does, however, have a lower bound. We previously defined the random variable X as non-negative, so that $E(X \mid X < t)$ cannot go below 0. $E(X \mid X \geq t)$ is also bound, with t as its limit.

Our lower bound for $E(X)$ is, therefore, given by, $E(X) \geq 0 \times P(X < t) + t \times P(X \geq t)$, as such

$$E(X) \geq t \times P(X \geq t)$$

$$\text{or, Markov's inequality,} \quad P(X \geq t) \leq \frac{E(X)}{t}$$

Markov's inequality is stating that the probability that X is at least t has an upper bound, or in simpler words, the probability of being big is not that big.

Similarly, we can describe the relationship between the variance of X and the probability that X is some distance from the mean, μ , using Chebyshev's lemma, which was one of the earliest results in probability theory that holds for any distribution, presented in 1867.

$$P(|X - \mu| \geq t) \leq \frac{\text{Var}(X)}{t^2}, \quad \text{Chebyshev's lemma}$$

Importantly, Chebyshev's inequality can be used to bound the error of a statistical estimator. For example, if we are aware of the distribution of our error, and want it to be close to 0 or the mean, μ , we can say that the probability of a big error is insignificant enough to discard.

Part 6: Brownian Motion, Symmetric Random Walk, and Martingales

In the late 17th century, Jacob Bernoulli performed work that is now recognised as the Bernoulli trials. He proposed flipping a coin a large enough number of times in order to demonstrate the law of large numbers, which tells us that the proportion of heads from all the flips will converge to the expected value, $\frac{1}{2}$. This is a stochastic process, and is, thus, defined by three key features:

- Each outcome of each step is random.
- The process evolves with respect to a continuous or discrete index, such as time, or flips of a coin.
- The process has a set of possible outcomes.

Let the outcomes of each flip be ω .

$$\omega = \omega_1 \omega_2 \omega_3 \dots$$

We are constantly flipping the coin, so that all ω have a result that is either heads or tails. For example, ω_1 is heads, ω_2 is tails, etc.

Suppose another scenario, in which each flip is denoted by ρ .

$$\rho = \rho_1 \rho_2 \rho_3 \dots$$

We can call each of these scenarios sample paths, and construct a random variable X , which takes the value 1 if ω is heads, and -1 if ω is tails. Now, using this variable we can mathematically define a stochastic process, M_k .

$$M_k = \sum_1^k X \quad \text{where } M \text{ begins at } 0.$$

We are, essentially, going through our outcomes of ω , and summing all the values for X , in k flips of the coin, where k is a positive integer.

Let's take an example. If our outcomes for ω are H (heads) H T (tails) H T T H T H, we can plot this stochastic process graphically.

figure 6.1

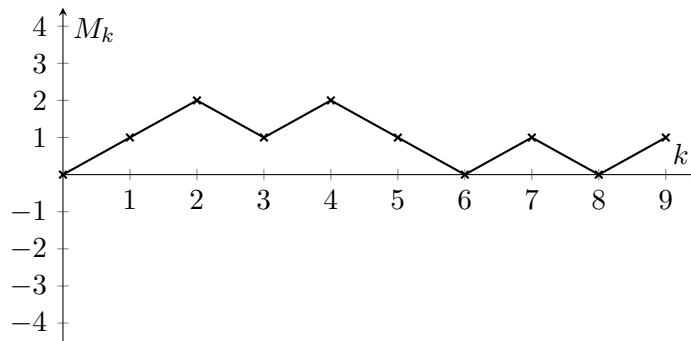


Figure 6.1 shows a graphical representation of our stochastic process, which we call a symmetric random walk.

Our symmetric random walk is an example of a discrete stochastic process, as we are iterating through individual flips of a coin. Brownian motion behaves in a similar way, however, in a continuous setting.

In order to model processes over time, we need to convert to the continuous space. Let us, once again, construct our probability space.

(Ω, F, P) where Ω is the sample space, F is a collection of subsets of Ω and P is the probability measure.

Let $W(t)$ be a stochastic process/function with time, $t \geq 0$ and $W(0) = 0$.

Properties of Brownian motion:

- It is important that each increment of W is independent, meaning it is not influenced by previous outcomes of W .

We should also note that our previous understanding of a random variable was limited. When discussing Markov's inequality, we referred to wage as a random variable, and considered it as the outcome of a random event (the wage of a random person in the population). We can now more accurately describe a random variable as a function that maps an event from the probability space, (Ω, F, P) , to a real number. Returning to independence, $W(t_n) - W(t_{n-1})$ describes an increment in the random variable W , so that for all n , these results are independent.

- Each increment also follows a normal distribution: $W(t_n) - W(t_{n-1}) \sim N(0, t_n - t_{n-1})$

From our discovery of the law of total expectation, we derived the result:

$$E(X | X < t) \times P(X < t) + E(X | X \geq t) \times P(X \geq t) = E(X)$$

For the random variable X , defined earlier, we can say that, since X is discrete, the expected value of X is

$$\sum x_i p_i = -1 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0$$

This property that the mean of the random variable is 0 holds true in the continuous space as well, and implies that Brownian motion has no tendency to drift towards one direction or the other.

The same is true for variance. For our discrete random variable X , $t_n - t_{n-1}$ would have always yielded an absolute value of 1, as evident in *figure 6.1*, where any increment by 1 in k causes M_k value to change by 1 too. We can calculate this to be the variance, using the standard result,

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 1 - 0 = 1$$

The variance, $t_n - t_{n-1}$, tells us that the larger the difference in successive measurements, the greater the variability of $W(t)$ is.

A mean of 0, along with the condition that $E(W(t)) < \infty$, tells us that Brownian motion has zero drift/is constant. A stochastic process with such a property can be referred to as a martingale, which can now help us consider how to predict events in the future given some information. If we know all the information about $W(t)$ up to time, t_n , and we want to predict $W(t)$ at time, t_m , then, mathematically, we want,

$E(W(t_m) | F_n)$ where F_n is all the information about $W(t)$ up to time, t_n , known as the filtration.

Since a martingale has zero drift, the best prediction for $W(t)$ at future time, t_m , is simply $W(t_n)$.

$$E(W(t_m) | F_n) = W(t_n)$$

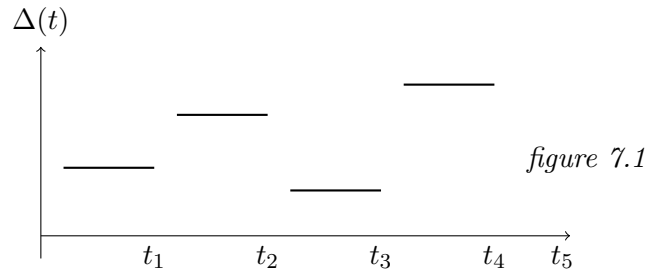
Part 7: Itô Integrals

In this section, we will begin to apply our knowledge of probability theory and Brownian motion to a more formal context in which we discuss trading, stock markets and profits and losses.

Let us define a stochastic process $\Delta(t)$, as the number of stocks being held by an individual at time, t . If we can consider this process from time 0 to time T , we can partition our time scale into n intervals, so that,

$$t_0 = 0 < t_1 < t_2 < t_3 \cdots < t_n = T$$

If $\Delta(t)$ is a simple process, so that between every interval $[t_j, t_{j+1})$, $\Delta(t)$ is constant, then it is also a step function and can be represented graphically, as in *figure 7.1*.



In the context of profits and losses, we will define another stochastic process $I(t)$ as the gain made by trading at time t , and $W(t)$ as the price per unit.

Suppose an interval $t_0 \leq t \leq t_1$. We can say that $I(t) = \Delta(t_1)W(t) - \Delta(t_0)W(t_0)$. This simply means the difference between the value of all the stocks at t and the value of the stocks at t_0 , assuming the stocks were sold at t .

Now, if we have an interval $t_1 \leq t \leq t_2$, then we must also consider the number of stocks held at t_0 .

$$I(t) = \Delta(t_0)W(t_0) + \Delta(t_1)W(t) - \Delta(t_1)W(t_1)$$

The first two terms describe the value of everything we have sold so far, and the last term describes the value of what we have bought.

Let us now generalise for an interval $t_k \leq t \leq t_{k+1}$. To consider everything that has happened before t_k , we need to sum the value of stocks bought and sold from t_0 to t_{k-1} ,

$$\sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))$$

We now need to add the gain in the interval $t_k \leq t \leq t_{k+1}$, which is $\Delta(t_k)(W(t) - W(t_k))$. All together, we have that

$$I(t) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k))$$

We call $I(t)$ an Itô integral and write the expression above as $\int_0^t \Delta(t) dW(t)$. This is not to be confused with standard integrals which are used to find the area under a curve. It is simply a shorthand form of expressing the stochastic process $I(t)$.

Itô integrals are the work of Japanese mathematician, Kiyoshi Itô, in 1944. Itô integration is a technique fundamental to pricing options and derivatives in quantitative finance. In the 1970s, Fischer Black, Myron Scholes and Robert Merton used Itô calculus to develop the Black-Scholes option pricing model.

Part 8: Options Pricing

To understand the Black-Scholes options pricing model, we first need to define an option. An option is a financial contract that gives the buyer the opportunity to buy a stock at a certain price, within a specific time period. For instance, if the price of one share of Google is £20, I could pay 50p for the option to buy a share at £22 within a 30-day period. If the stock price appreciates within the 30-day period, and is now at £30, then I can buy it at £22 and immediately sell the option at £30, to make a profit of £7.50. The upfront cost that is paid by the buyer is referred to as the option price, and the share price which the option outlines is the strike price.

The Black-Scholes options pricing model states that, under the assumption of a 1-year expiration date, and 0 interest rates, the option price, C , is

$$C = S N\left(\frac{\ln\left(\frac{S}{K}\right) + \frac{\sigma^2}{2}}{\sigma}\right) - K N\left(\frac{\ln\left(\frac{S}{K}\right) - \frac{\sigma^2}{2}}{\sigma}\right)$$

Indeed this equation looks daunting, however we can break it down easily to understand each component individually.

$\frac{S}{K}$ is simply a ratio between the underlying share price, S , and the strike price, K . Taking the natural log of this ratio calibrates it with respect to 0, such that if the share price is greater than the strike price, then $\ln\left(\frac{S}{K}\right)$ is positive, and if the share price is less than the strike price, then $\ln\left(\frac{S}{K}\right)$ is negative.

σ refers to the volatility of the stock. If we consider what happens as σ approaches 0, and $S = K$, we can deduce that the expression inside the normal function tends to 0. If $S > K$, the expression will tend to infinity, and if $S < K$, it will tend to negative infinity.

The function, $N()$, is the cumulative normal distribution, with a mean, $\mu = 0$. Thus, when the expression $\frac{\ln(\frac{S}{K}) + \frac{\sigma^2}{2}}{\sigma} = 0$, $N\left(\frac{\ln(\frac{S}{K}) + \frac{\sigma^2}{2}}{\sigma}\right) = 0.5$. This tells us how sensitive the option price is relative to the underlying stock price, and is called delta by option traders. If the volatility is greater than 0, then for every change in stock price by £1, the price of the option will change by a greater amount than if the volatility was 0.

If we now introduce the expiration time into the equation, we have that,

$$C = S N\left(\frac{\ln(\frac{S}{K}) + T\frac{\sigma^2}{2}}{\sigma\sqrt{T}}\right) - K N\left(\frac{\ln(\frac{S}{K}) - T\frac{\sigma^2}{2}}{\sigma\sqrt{T}}\right)$$

Introducing the expiration time of the option indicates that over a larger time period, the effects of volatility are heightened, since we allow for more variation in the underlying stock price.

Therefore, the left-hand side of the equation describes, given a stock price, S , what is the relative payout of the stock price against delta. The right-hand side is similar. It tells us what we are giving up by owning the option.

$N\left(\frac{\ln(\frac{S}{K}) - T\frac{\sigma^2}{2}}{\sigma\sqrt{T}}\right)$ tells us the sensitivity of the option price relative to the strike price K .

Clearly, I have not mentioned stochastics or Itô integrals when discussing the Black-Scholes formula. In actuality, these do not come up directly in the model, but the use of stochastic differential equations is necessary when modelling the randomness of the underlying asset price.

Part 9: Overview

Overall, we have explored how probability theory and some key ideas of Brownian motion can be used as financial modelling techniques, especially when trying to model randomness. Many of the techniques that I have discussed are simplifications of the actual methods used and many also have their limits. For example, using stochastic calculus to model stock prices is ineffective when market conditions are heavily determined by macroeconomic factors, which make precise predictions impossible. The Black-Scholes model also fails to consider changes in volatility within the expiration time and behavioural factors such as herd mentality. Stochastics can provide a powerful theoretical framework for modelling market behaviour, however, struggles when applied to real-world scenarios. This has led to the development of other, more complex models, such as jump diffusion models which account for discrete jumps in asset prices in order to better capture extreme market events, such as crashes or earnings surprises.